

Marina Rice, M.S Bioinformatics

California | 424-319-6095 | mrice183@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Bioinformatics M.S. student and clinical genomics professional with experience supporting high-volume genomic testing and a background spanning clinical genomics, clinical research, and biological data analysis. Project experience includes Python- and R-based work in functional gene annotation, spatial biology visualization, transcriptomic research, and reproducible computational workflows. Brings strengths in biological interpretation, data quality, and scientific visualization while continuing to develop programming and data analysis skills. Seeking opportunities in bioinformatics, computational biology, and genomic data analysis that combine biological context with reproducible computational approaches.

SKILLS

Programming & Data: Python • R • Pandas • SQL • Bash/Linux • Git/GitHub

Scientific Analysis: DESeq2 • RNA-seq • Spatial biological data • Statistical analysis • Biological interpretation

Visualization & Applications: Streamlit • Plotly • Interactive biological data visualization • Data exploration

Genomics & Quality: WES/WGS workflows • BLAST • HMMER • TMHMM • QC • Data validation • Reproducible documentation

PROJECTS

[Neurodegenerative Disorder Biomarker Discovery Capstone](#) - 1st Place Team

CFDE DeCoDE Institute | Team July 2026

- Collaborated on a team project investigating shared extracellular RNA biomarkers across neurodegenerative and cognitive disorders using plasma small RNA-seq data.
- Gained hands-on exposure to DESeq2 by independently testing differential-expression analysis on one disease population while contributing biological research and interpretation to the team's final analysis.
- Supported interpretation and communication of quantitative findings in a multidisciplinary project investigating shared biomarkers across neurodegenerative and cognitive disorders.

[Spatial Transcriptomics Visualization Dashboard](#) - 3rd Place

CFDE Data Visualization Challenge | June 2026

- Developed a Streamlit application using Python, Pandas, and Plotly to explore high-dimensional spatial biological data from human lymph-node tissue.
- Built interactive filtering and visualization features that made complex spatial patterns easier to explore and connect with biological contexts.

[Rule-Based Gene Annotation & Sorting Pipeline](#)

UCSD Extended Studies - Bioinformatics | May 2026

- Developed a Python workflow that parsed BLAST, HMMER, and TMHMM outputs and applied a tiered evidence hierarchy to assign putative gene functions.
 - Integrated multiple computational evidence sources into standardized annotation outputs, demonstrating data parsing, structured logic, and reproducible workflow design.
-

EXPERIENCE

GeneDx | Genetic Counseling Assistant II | Jul 2025 – Present

- Process approximately 1,800 clinical accessions per month across high-volume genomic workflows while maintaining an average error rate of approximately 0.5%.
- Perform quality-control checks, resolve workflow discrepancies, and maintain accurate records while meeting production and turnaround requirements.
- Support Rapid genomic testing workflows requiring accelerated turnaround times, proactive issue resolution, and end-to-end case coordination from accession review through downstream case readiness.

CenExel Clinical Research | Research Assistant | Jan 2025 – Jul 2025

- Maintained and reconciled clinical research data across regulated Phase I-III study workflows, supporting data quality, documentation, and query resolution.
- Used Excel and clinical research systems to track outstanding work and collaborated with coordinators, data managers, and CRAs on data and workflow issues.

EDUCATION

University of Maryland Global Campus | M.S. Bioinformatics | Expected August 2028

Relevant Coursework: Intro to Bioinformatics

UC San Diego Extended Studies | Bioinformatics Certificate | Expected January 2027

Relevant Coursework: Medical Microbiology, Bioinformatics Fundamentals, Genomic Sequencing Technologies, Processing Actionable Data in Genomics

California State University, Long Beach | B.S. Biology